

UKRAINIAN CATHOLIC UNIVERSITY

BACHELOR THESIS

Autonomous Ground Navigation and Dynamic Obstacle Avoidance for Husky Robot in Pedestrian Areas

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*A thesis submitted in fulfillment of the requirements
for the degree of Bachelor of Science*

in the

Department of Computer Sciences and Information Technologies
Faculty of Applied Sciences



Lviv 2026

“TODO? Thanks to my solid academic training, today I can write hundreds of words on virtually any topic without possessing a shred of information, which is how I got a good job in journalism.”

Dave Barry

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Husky Robot in Pedestrian Areas**

by Radomyr HUSIEV

Abstract

TODO

Acknowledgements

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List of Abbreviations

TD To Do

Physical Constants

TODO? $c_0 = 2.997\,924\,58 \times 10^8 \text{ m s}^{-1}$ (exact)

List of Symbols

t TODO rad

For/Dedicated to/To my... TODO?

Chapter 1

Introduction

1.1 Background and Motivation

Ground vehicles – very important field; especially for tedious, far-away, or dangerous tasks.

It is often infeasible to manually control such vehicles due to their sheer number and task volume. Physical barriers like distance or signal interference mean manual control can fail. Therefore, autonomous systems are a necessity for the robot to complete its task and navigate to desired locations independently.

Outdoors, there are challenges of an unpredictable environment: unexpected/moving obstacles, surrounding agents (like other vehicles or pedestrians) don't cooperate, uneven terrain, etc.

Today, there are ready-made solutions for autonomous navigation (like default Nav2 in ROS). However, modern outdoor systems often rely on heavy sensors (3D LiDAR, RGB-D cameras) and algorithms to process this data (Visual SLAM, MPPI). These require high computational resources, lots of RAM, and sometimes GPUs. This is expensive, it drains battery and not always possible on simple low-cost robots.

Therefore, there is a motivation to see what can be accomplished using a minimal, lightweight sensor suite. By avoiding heavy computations, we can create a more accessible and energy-efficient system.

Instead of calculating a map from scratch using SLAM, we can use existing, semantic data like OpenStreetMap (OSM) for global routing. Instead of 3D point clouds or image processing, we can use a simple 2D LiDAR to handle immediate, dynamic obstacles (like pedestrians or trees). Instead of visual odometry, we can rely on standard GPS fused with an IMU and wheel odometry.

Together, these lightweight components theoretically provide all the necessary data for a robot to navigate (albeit in lower fidelity and density, higher sparsity). This thesis aims to prove that such a minimal setup is actually viable in the real world.

The setup includes a ground differentially driven robot outdoors in a park environment, specifically on pedestrian paths in a known (topologically, but not dynamically) environment. It should navigate autonomously from a start point to a target point without human intervention.

1.2 Objectives

- Path planning – find a global route (using graph search algorithms like A*) based on environment map (OpenStreetMap XML data).
- Obstacle detection – process 2D LiDAR data to find obstacles.
- Local navigation – combine spatial localization (GPS and IMU+Wheel Odometry – sensor fusion via Extended Kalman Filter) and local path planning (using

an optimization algorithm Timed Elastic Band) to avoid obstacles and follow the global path.

Conducted real-world experiments in public environment (Stryyskyi Park). Also in simulation – more times here. Evaluated the system’s success rate, safety, and travel time. Provided a baseline comparison against the standard Nav2 implementation.

1.3 Structure Of The Thesis

Chapter 2

Related Work

Chapter 3

Theoretical Background

Chapter 4

Proposed Solution

4.1 Problem Formulation

Chapter 5

Experiments and Results

Chapter 6

Conclusions

Appendix A

Appendix